

Appendix A

Background on alphabets

People sometimes tell me that they were fine with math “until the numbers became letters”. We use letters to represent unknown quantities, and in abstract math basically everything is an unknown quantity so we end up needing rather a lot of letters. We also often want to use different types of letter for different things, to help us maintain some intuition about what is going on. Here are some ways in which I typically use roman letters throughout this book. These are not hard-and-fast rules.

- natural numbers: n, m, k
- elements of sets or objects of categories: a, b, c, x, y, z
- functions or morphisms: f, g, p, q, s, t
- sets: A, B, C, X
- functors: F, G, H, K
- specific sets of numbers in standard notation: blackboard bold $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
- small categories: blackboard bold $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$
- locally small categories: curly $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$

Once we find ourselves running out of roman letters we branch out into Greek letters. It’s not a bad idea to look them up and familiarize yourself with them although in this book really the only ones I use are the first four (shown on the right) for natural transformations, and ω (omega) for countable infinity.

α	alpha
β	beta
γ	gamma
δ	delta

The next letter is epsilon (ϵ or ε) and is classically used in analysis to torture undergraduates in calculus class, or rather, to construct rigorous proofs in calculus using the idea of an unknown distance which can be arbitrarily small but is never 0. Thus proofs in calculus often begin “Let $\varepsilon > 0$ ”, so much so that I have been known to tell students that if all else fails they can write that and I will give them one point. This also gives rise to a joke which I find ludicrously funny: “Let ε be a large negative integer.” This is funny because once you’re so incredibly used to ε being an arbitrarily small positive real number, the idea of ε representing a large negative integer is, well, inexplicably hilarious.